

Week	Humanoid Capability	Real-World Mission	Mathematics Mapping	EDC Mapping	C Programming Mapping	AI Integration	Mini Project	Faculty Coordination	Student Narrative	GitHub Deliverables
1	Robot Brain	Teach robot numbers	Matrix Basics	Binary Representation	Variables, Arrays	Data Representation	Matrix Calculator	Math + C + EDC	"Today we build the robot's brain."	README, Matrix Calculator
2	Robot Memory	Store information	Matrix Operations	Memory Organization	2D Arrays	AI Memory	Memory Storage Simulator	Math + C	"Every robot remembers through memory."	Source Code + Documentation
3	Robot Eyes	Understand images	Matrix Transformation	Light Sensor Basics	2D Arrays	Computer Vision	Pixel Image Viewer	All Faculty	"A robot sees using cameras."	Image Matrix Program
4	Robot Skin	Sense temperature	Functions & Graphs	Diodes, Sensors	Functions	Sensor AI	Temperature Monitor	Math + EDC	"The robot learns to feel heat."	Sensor Documentation
5	Robot Hearing	Capture sound	Graph Analysis	Microphone Signal	Arrays	Speech AI	Sound Level Analyzer	EDC + C	"The robot hears your voice."	Audio Processing Program
6	Robot Decision Unit	Make choices	Mean Value Concepts	Signal Threshold	if-else, switch	Rule-Based AI	Smart Decision Engine	Math + C	"Every robot must decide."	Decision Logic Code
7	Robot Hands	Pick an object	Coordinate Basics	Servo Motors	Loops	Motion Planning	Robotic Arm Simulator	All Faculty	"Today our robot moves its hand."	Motion Program
8	Robot Legs	Walking	Calculus Basics	DC Motor	Loops & Functions	Path Planning	Walking Simulator	Math + EDC + C	"Teach the robot to walk."	Walking Simulation
9	Robot Balance	Stand steadily	Derivatives	Motor Control	Functions	Stability Control	Self-Balance Simulator	All Faculty	"How does a robot avoid falling?"	Balance Report
10	Robot Face	Display emotions	Matrices	LED/LCD	Strings	Human Interaction	Emoji Display System	EDC + C	"The robot expresses emotions."	UI Prototype
11	Robot Voice	Speak	Functions	Speaker Circuit	Strings	Text-to-Speech	Robot Voice Generator	All Faculty	"Your robot speaks today."	Demo Video
12	Robot Communication	Exchange data	Matrices	Communication Basics	Functions	AI Chat	Local Chat Interface	C Faculty	"Robots talk with humans."	Chat Program
13	Robot Navigation	Know direction	Vectors	Distance Sensor	Arrays	Navigation AI	Path Finder	Math + EDC	"Navigate through a room."	Navigation Code
14	Robot Mapping	Build environment map	Jacobians	Ultrasonic Sensor	Structures	SLAM Introduction	Room Mapping	All Faculty	"Robot creates its own map."	Map Output
15	Robot Touch	Object Detection	Partial Derivatives	IR Sensor	Structures	Object Recognition	Smart Touch Sensor	All Faculty	"Robot identifies objects."	Detection Demo
16	Robot Vision AI	Detect objects	Matrix Algebra	Camera Module	File Handling	Object Detection	Image Classifier	Math + C	"Robot identifies people."	Image Recognition
17	Robot Health	Self Diagnostics	Optimization	Voltage Monitoring	Functions	Predictive Maintenance	Health Dashboard	EDC + C	"Robot checks itself."	Dashboard
18	Robot Energy	Battery Management	Integration	Power Electronics	Files	Battery AI	Energy Monitor	Math + EDC	"Robot manages its battery."	Report
19	Robot Learning	Learn from experience	Eigen Values	Signal Processing	Pointers	Machine Learning	Pattern Learner	Math + C	"Robot starts learning."	ML Prototype
20	Robot Intelligence	Connect modules	Multi-variable Functions	Sensor Integration	Structures	AI Decision Making	AI Controller	All Faculty	"Brain controls the body."	Integrated Code
21	Robot Assistant	Help humans	Optimization	Human Interface	Functions	Personal Assistant	Smart Assistant	All Faculty	"Robot becomes useful."	Demo
22	Robot Security	Safe operation	Matrix Security	Protection Circuits	File Security	AI Safety	Access Controller	EDC + C	"Robot protects itself."	Security Module
23	Robot Collaboration	Work with others	Vector Spaces	Communication Bus	Files	Multi-Agent AI	Robot Team Simulation	All Faculty	"Robots cooperate."	Collaboration Demo
24	Robot Industry	Factory Robot	Linear Systems	PLC Concepts	Modular Programming	Industrial AI	Pick-and-Place Simulator	All Faculty	"Become an Industry 4.0 engineer."	Project Report
25	Robot Healthcare	Medical Assistant	Optimization	Biomedical Sensors	Structures	Healthcare AI	Patient Monitoring	All Faculty	"Robot saves lives."	Healthcare Demo
26	Robot Agriculture	Smart Farming	Coordinate Systems	Soil Sensors	Files	Agri AI	Smart Irrigation	All Faculty	"Robot helps farmers."	Agri Report
27	Robot Disaster Response	Rescue Robot	Vector Calculus	Gas Sensors	Recursion	Emergency AI	Rescue Simulator	All Faculty	"Robot saves people."	Rescue Demo
28	Complete Integration	Connect all modules	Revision	Revision	Revision	System AI	Integrated Humanoid	All Faculty Together	"Everything comes together."	Integrated Repository
29	Testing & Validation	Debug system	Mathematical Validation	Circuit Testing	Debugging	AI Evaluation	System Testing	All Faculty	"Engineers validate before deployment."	Testing Report
30	Final Showcase	Deploy Humanoid	Complete Revision	Complete Revision	Complete Revision	Industry Demo	KIET AI Humanoid Expo	Faculty Panel + Industry	"You are no longer students—you are AI Physical Systems Engineers."	Final GitHub Repository + Demo + PPT